



Complex Numbers · Mark Scheme

Core Pure 2 Chapter 1: Complex Numbers. M = method mark A = accuracy mark (depends on M) B = independent mark ft = follow through

Section A – De Moivre's theorem and powers

- 1 $\cos \frac{6\pi}{9} + i \sin \frac{6\pi}{9} = \cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3}$ M1; $= -\frac{1}{2} + \frac{\sqrt{3}}{2}i$ A1.
- 2 Numerator $\cos 12\theta + i \sin 12\theta$ and $\cos \theta - i \sin \theta = \cos(-\theta) + i \sin(-\theta)$, so denominator $= \cos(-2\theta) + i \sin(-2\theta)$ M1; $\cos 14\theta + i \sin 14\theta$ A1.
- 3 $1 + \sqrt{3}i = 2(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3})$ B1; $2^9(\cos 3\pi + i \sin 3\pi)$ M1; $= -512$ A1.
- 4 $\sqrt{3} + i = 2(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6})$ and $1 - i = \sqrt{2}(\cos -\frac{\pi}{4} + i \sin -\frac{\pi}{4})$ B1; numerator $128(\cos \frac{7\pi}{6} + i \sin \frac{7\pi}{6})$, denominator $4(\cos(-\pi) + i \sin(-\pi)) = -4$ M1; quotient $32(\cos \frac{13\pi}{6} + i \sin \frac{13\pi}{6}) = 32(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6})$ M1; $= 16\sqrt{3} + 16i$ A1.
- 5(a) $n = 1$: LHS $= \cos \theta + i \sin \theta =$ RHS, so true for $n = 1$ B1; assume true for $n = k$: $(\cos \theta + i \sin \theta)^{k+1} = (\cos k\theta + i \sin k\theta)(\cos \theta + i \sin \theta)$ M1; $= (\cos k\theta \cos \theta - \sin k\theta \sin \theta) + i(\sin k\theta \cos \theta + \cos k\theta \sin \theta)$ M1; $= \cos(k+1)\theta + i \sin(k+1)\theta$ using the compound angle formulae A1; so if true for $n = k$ it is true for $n = k+1$; true for $n = 1$, hence true for all $n \in \mathbb{Z}^+$ A1.
- 5(b) $(\sqrt{3} + i)^n = 2^n(\cos \frac{n\pi}{6} + i \sin \frac{n\pi}{6})$ M1; real and positive when $\frac{n\pi}{6} = 2\pi m$, so smallest $n = 12$ A1; value $2^{12} = 4096$ A1.

Section B – Trigonometric identities

- 6 $z^n = \cos n\theta + i \sin n\theta$ and $z^{-n} = \cos(-n\theta) + i \sin(-n\theta) = \cos n\theta - i \sin n\theta$ M1; adding gives $2 \cos n\theta$ A1.
- 7 $\cos 3\theta + i \sin 3\theta = (c + is)^3$, where $c = \cos \theta$, $s = \sin \theta$ M1; real part: $\cos 3\theta = c^3 - 3cs^2$ A1; $= c^3 - 3c(1 - c^2) = 4c^3 - 3c$ A1.
- 8 $(z + \frac{1}{z})^4 = 16 \cos^4 \theta$ and $= z^4 + 4z^2 + 6 + 4z^{-2} + z^{-4}$ M1; $= 2 \cos 4\theta + 8 \cos 2\theta + 6$ M1; $\cos^4 \theta = \frac{1}{8} \cos 4\theta + \frac{1}{2} \cos 2\theta + \frac{3}{8}$ A1.
- 9 $(c + is)^3 = c^3 - 3cs^2 + i(3c^2s - s^3)$ M1; $\tan 3\theta = \frac{3c^2s - s^3}{c^3 - 3cs^2}$ A1; divide numerator and denominator by c^3 M1; $= \frac{3 \tan \theta - \tan^3 \theta}{1 - 3 \tan^2 \theta}$ A1.
- 10(a) $\cos 5\theta = \operatorname{Re}(c + is)^5 = c^5 - 10c^3s^2 + 5cs^4$ M1 A1; substitute $s^2 = 1 - c^2$: $c^5 - 10c^3(1 - c^2) + 5c(1 - 2c^2 + c^4)$ M1; $= 16c^5 - 20c^3 + 5c$ A1.
- 10(b) Let $x = \cos \theta$: $2 \cos 5\theta - 1 = 0$, so $\cos 5\theta = \frac{1}{2}$ M1; $5\theta = \frac{\pi}{3}, \frac{5\pi}{3}, \frac{7\pi}{3}, \frac{11\pi}{3}, \frac{13\pi}{3}$, so $\theta = \frac{\pi}{15}, \frac{\pi}{3}, \frac{7\pi}{15}, \frac{11\pi}{15}, \frac{13\pi}{15}$ M1 A1; $x = 0.978, 0.5, 0.105, -0.669, -0.914$ A1.
- 10(c) Product of the five roots $= \frac{1}{32}$ M1; one root is $\cos \frac{\pi}{3} = \frac{1}{2}$, so the product $= \frac{1}{32} \div \frac{1}{2} = \frac{1}{16}$ A1.

Section C – n th roots

- 11 $r = \sqrt{4 + 12} = 4$ B1; $\theta = \frac{2\pi}{3}$, so $4e^{\frac{2\pi}{3}i}$ B1.
- 12 $27i = 27e^{\frac{\pi}{2}i}$, so $z = 3e^{i(\frac{\pi}{6} + \frac{2k\pi}{3})}$ M1; $3e^{\frac{\pi}{6}i}, 3e^{\frac{5\pi}{6}i}$ A1; $3e^{-\frac{\pi}{2}i}$ A1.
- 13 $-16 = 16e^{\pi i}$, so $z = 2e^{i(\frac{\pi}{4} + \frac{k\pi}{2})}$ M1; arguments $\pm \frac{\pi}{4}, \pm \frac{3\pi}{4}$ A1; $z = \sqrt{2} + \sqrt{2}i, -\sqrt{2} + \sqrt{2}i, -\sqrt{2} - \sqrt{2}i, \sqrt{2} - \sqrt{2}i$ A1.
- 14 $16 + 16\sqrt{3}i = 32e^{\frac{\pi}{3}i}$ B1; $z = 2e^{i(\frac{\pi}{15} + \frac{2k\pi}{5})}$ M1; $2e^{\frac{\pi}{15}i}, 2e^{\frac{7\pi}{15}i}, 2e^{\frac{13\pi}{15}i}$ A1; $2e^{-\frac{11\pi}{15}i}, 2e^{-\frac{\pi}{15}i}$ A1.
- 15(a) $r = 8$ B1; $\theta = -\frac{\pi}{4}$, so $8e^{-\frac{\pi}{4}i}$ B1.
- 15(b) $z = 8^{\frac{1}{3}}e^{i(-\frac{\pi}{4} + \frac{2k\pi}{3})}$ M1; modulus 2 B1; $2e^{-\frac{\pi}{12}i}, 2e^{\frac{7\pi}{12}i}$ A1; $2e^{-\frac{3\pi}{4}i}$ A1.



15(c) Three points on a circle of radius 2 centred at O $B1$; equally spaced at angles $\frac{2\pi}{3}$ apart, in the correct positions $B1$.

15(d) Equilateral triangle made of three triangles with sides 2, 2 and angle $\frac{2\pi}{3}$: area $= 3 \times \frac{1}{2} \times 2 \times 2 \times \sin \frac{2\pi}{3}$ $M1$; $= 3\sqrt{3}$ $A1$.

Section D – Roots of unity

16 $z = e^{\frac{2k\pi}{3}i}$, $k = 0, 1, 2$ $M1$; $1, -\frac{1}{2} + \frac{\sqrt{3}}{2}i, -\frac{1}{2} - \frac{\sqrt{3}}{2}i$ $A1$.

17 Geometric series: $1 + \omega + \dots + \omega^4 = \frac{\omega^5 - 1}{\omega - 1}$ $M1$; $\omega^5 = e^{2\pi i} = 1$ and $\omega \neq 1$, so the sum is 0 $A1$.

18 $z = e^{\frac{2k\pi}{6}i} = e^{\frac{k\pi}{3}i}$ $M1$; $1, e^{\pm \frac{\pi}{3}i}, e^{\pm \frac{2\pi}{3}i}, e^{\pi i}$ $A1$; the vertices of a regular hexagon (centre O , inscribed in the unit circle) $B1$.

19 $1 + \omega + \omega^2 = 0$ $B1$; (a) $1 + \omega + \omega^2 + \omega^3 = 0 + 1 = 1$ $B1$; (b) $2 + 5\omega + 2\omega^2 = 2(1 + \omega + \omega^2) + 3\omega = 3\omega$ $M1$; $(3\omega)^6 = 729\omega^6 = 729$ $A1$.

20(a) $\sqrt{3} + i = 2e^{\frac{\pi}{6}i}$, so $c = 2^6 e^{\pi i}$ $M1$; $c = -64$ $A1$.

20(b) Multiply successively by $e^{\frac{\pi}{3}i}$ (rotation by $\frac{\pi}{3}$) $M1$; $B = 2i$, $C = -\sqrt{3} + i$ $A1$; $D = -\sqrt{3} - i$, $E = -2i$, $F = \sqrt{3} - i$ $A1$.

20(c) Six equilateral triangles of side 2: $6 \times \frac{1}{2} \times 2^2 \times \sin \frac{\pi}{3}$ $M1$; $= 6\sqrt{3}$ $A1$.

20(d) 0, since the roots of $z^6 + 64 = 0$ have sum 0 (no z^5 term), or by symmetry $B1$.

Roar-level challenge

21 $(z - \frac{1}{z})^5 = (2i \sin \theta)^5 = 32i \sin^5 \theta$ $M1$; $= z^5 - 5z^3 + 10z - 10z^{-1} + 5z^{-3} - z^{-5}$ $A1$; $= 2i \sin 5\theta - 10i \sin 3\theta + 20i \sin \theta$,
so $\sin^5 \theta = \frac{1}{16} \sin 5\theta - \frac{5}{16} \sin 3\theta + \frac{5}{8} \sin \theta$ $A1$; $\int_0^{\frac{\pi}{2}} = [-\frac{1}{80} \cos 5\theta + \frac{5}{48} \cos 3\theta - \frac{5}{8} \cos \theta]_0^{\frac{\pi}{2}}$ $M1$; $= \frac{1}{80} - \frac{5}{48} + \frac{5}{8} = \frac{8}{15}$ $A1$.